

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTEEC401 - NETWORK THEORY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Four resistances $80\ \Omega$, $50\ \Omega$, $25\ \Omega$, and R are connected in parallel. Current through $25\ \Omega$ resistor is 4 A . Total current of the supply is 10 A . The value of R will be
(a) $66.66\ \Omega$ (b) $40.25\ \Omega$ (c) $36.36\ \Omega$ (d) $76.56\ \Omega$
- A network contains only an independent current source and resistors. If the values of all resistors are doubled, the value of the node voltages will
(a) become half (b) remain unchanged (c) become double (d) none of these
- The nodal method of circuit analysis is based on
(a) KVL and Ohm's law (b) KCL and Ohm's law (c) KCL and KVL (d) KCL, KVL and Ohm's law
- 4 A series RLC circuit has a resonance frequency of 1 kHz and a quality factor Q of 100 . If each of R , L and C is doubled from its original value, the new Q of the circuit is
(a) 25 (b) 50 (c) 100 (d) 200
- For a series resonant circuit at low frequency, circuit impedance is _____ and at high frequency circuit impedance is _____.
(a) capacitive, inductive (b) inductive, capacitive (c) resistive, inductive (d) capacitive, resistive
- 6 The coupling between two magnetically coupled coils is said to be ideal if the coefficient of coupling is
(a) zero (b) 0.5 (c) 1 (d) 2
- 7 A network has 7 nodes and 5 independent loops. The number of branches in the network is
(a) 13 (b) 12 (c) 11 (d) 10
- 8 A step function voltage is applied to an RLC series circuit having $R = 2\ \Omega$, $L = 1\text{ H}$ and $C = 1\text{ F}$. The transient current response of the circuit would be
(a) over damped (b) critically damped (c) under damped (d) none of these

9 When a unit-impulse voltage is applied to an inductor of 1 H, the energy supplied by the source is

- (a) ∞ (b) 1 J (c) 0.5 J (d) 0

10 The transfer function of a low-pass RC network is

- (a) $(RCs)(1 + RCs)$ (b) $\frac{1}{1 + RCs}$
 (c) $\frac{RCs}{1 + RCs}$ (d) $\frac{s}{1 + RCs}$

As the poles of a network shift away from the x axis, the response

- (a) remains constant (c) becomes more oscillating
 (b) becomes less oscillating (d) none

The h parameters of the circuit shown in Fig.

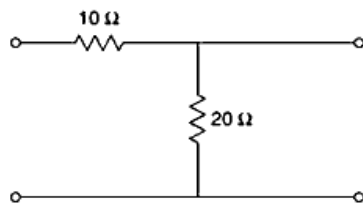
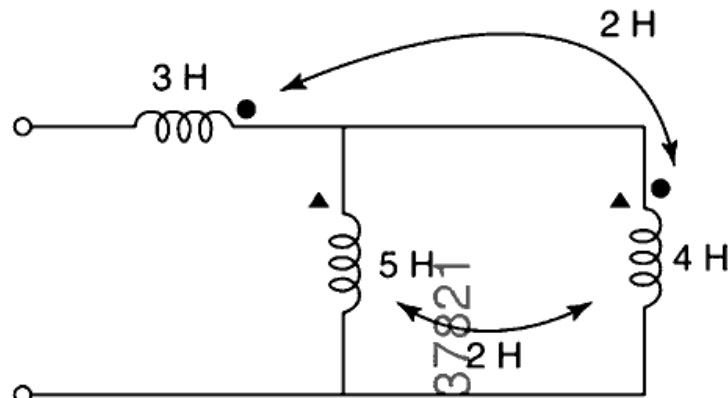


Fig. 13.156

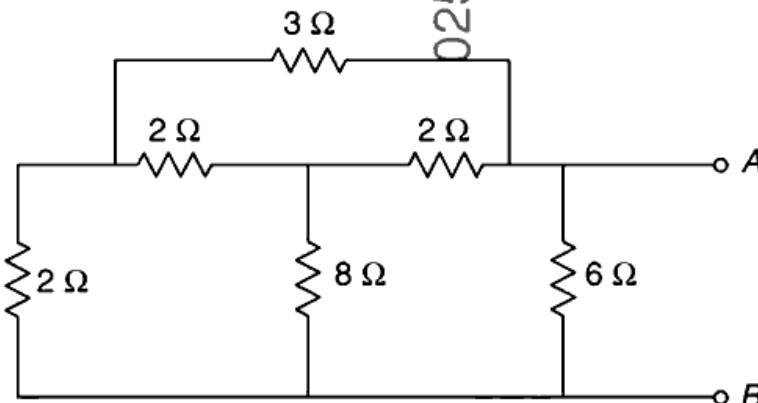
- (a) $\begin{bmatrix} 0.1 & 0.1 \\ -0.1 & 0.3 \end{bmatrix}$ (b) $\begin{bmatrix} 10 & -1 \\ 1 & 0.05 \end{bmatrix}$
 (c) $\begin{bmatrix} 30 & 20 \\ 20 & 20 \end{bmatrix}$ (d) $\begin{bmatrix} 10 & 1 \\ -1 & 0.05 \end{bmatrix}$

Q2. Solve the following.

Find the effective inductance of the network shown in Fig.



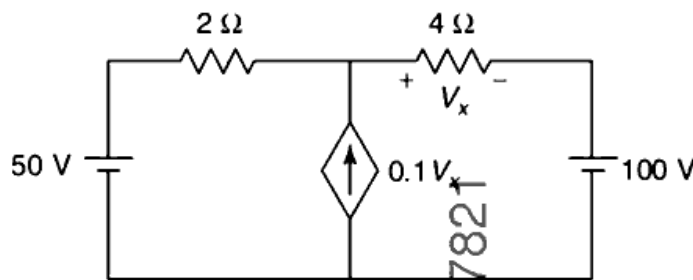
Find the equivalent resistance between A and B in the network.



Q3. Solve the following.

- A) State and Explain Tellegen's Theorem and Millman's theorem.
 B) Find voltage V_x using superposition theorem.

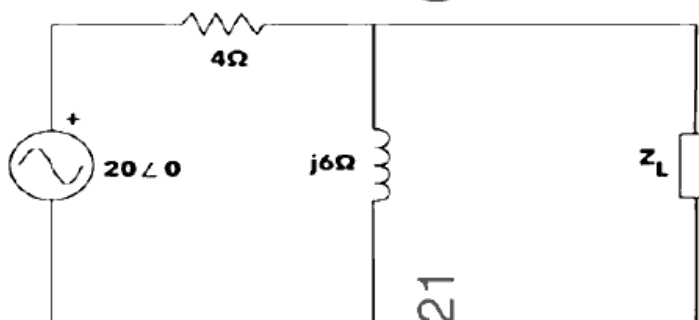
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Q4. Solve Any Two of the following.

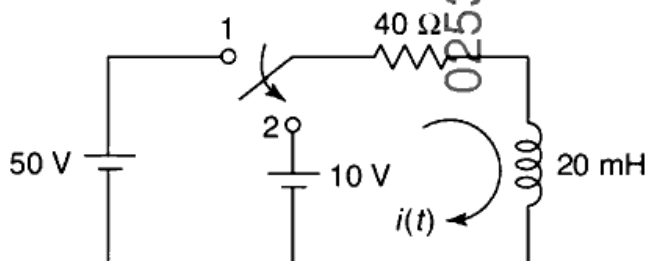
- Calculate value of Z_L and maximum power at Z_L for following network by using maximum power theorem

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The network of Fig. is under steady state with switch at the position 1. At $t = 0$, switch is moved to position 2. Find $i(t)$.

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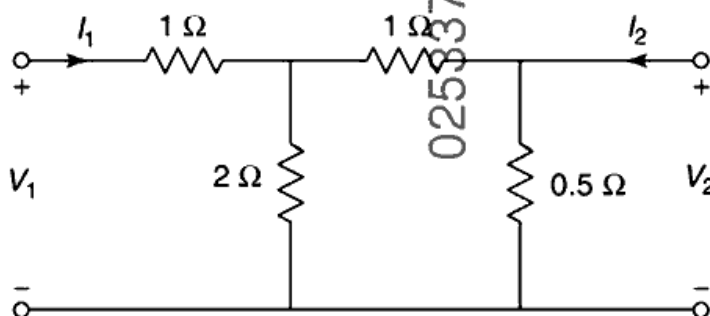
- C) Derive Transient and steady state response of R-L circuit to various with Unit step functions

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Q5. Solve Any Two of the following.

- Determine Z-parameters for the network shown in Fig.

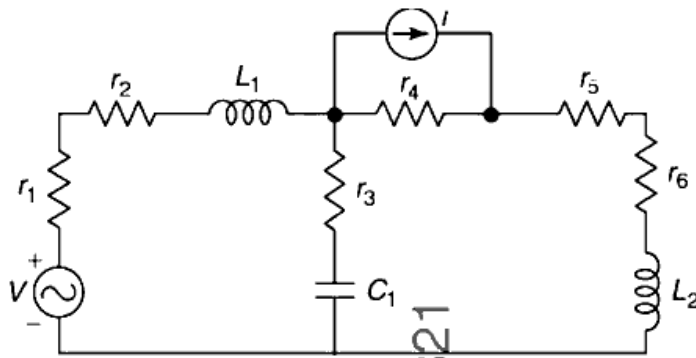
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- B) Explain with neat diagram Transmission Parameters

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- C) For the circuit shown in Fig. draw the oriented graph and write (a) incidence matrix, (b) tie-set matrix, and (c) cut-set matrix. 6



Q6. Solve Any Two of the following.

- A) Classify various types of filters. Derive transfer function of active High-pass filters and draw its characteristics. 6
 B) A series resonant circuit has an impedance of 500 Ω at resonant frequency. Cut-off frequencies are 10 kHz and 100 Hz. Determine (a) resonant frequency, (b) value of L, C, and (c) quality factor at resonant frequency. 6
 C) Explain the concept of parallel resonance in an RLC circuit. Describe the conditions necessary for parallel resonance to occur and discuss the behavior of the circuit at resonance. 6

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